

LIPSYNC.AI: A.I. Driven Lips and Tongue Animations Using Articulatory Phonetic Descriptors and FACS Blendshapes

Jara Alvarez Masso
jara@lipsync.ai
Emotech
United Kingdom

Alexandru Mihai Rogozea
alex@emotech.co
Emotech
United Kingdom

Dr. Saeid Mokaram
saeid.mokaram@gmail.com
Emotech
United Kingdom

Dr. Jan Medvesek
j@emotech.co
Emotech
United Kingdom

Yijun Yu
papaya4215@gmail.com
Emotech
United Kingdom

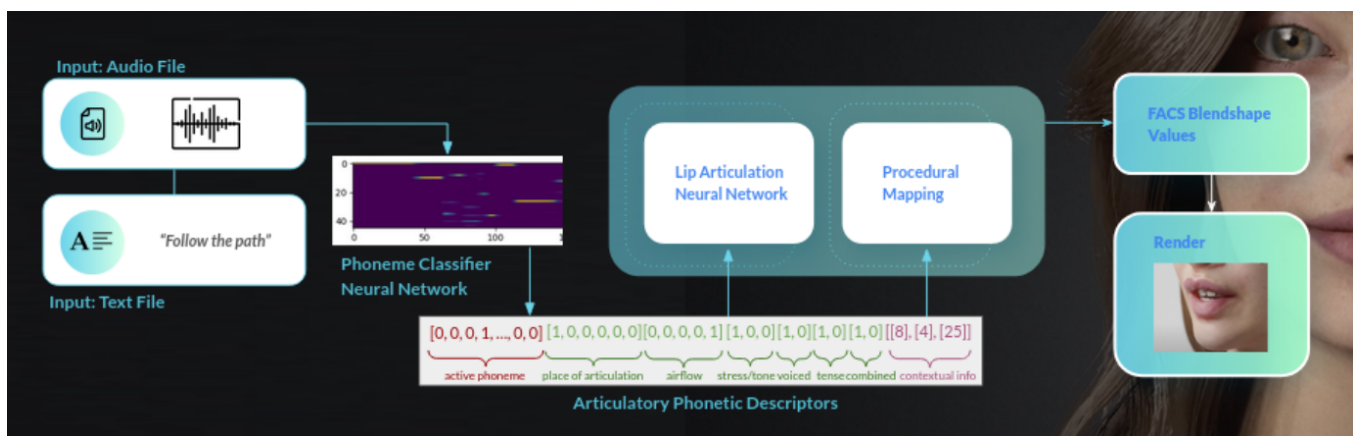


Figure 1: The flow of data through the modules of LIPSYNC.AI, highlighting the addition of the articulatory phonetic descriptors

ABSTRACT

We present a solution for generating realistic lips and tongue animations, using a novel hybrid method which makes use of both the advancements in deep learning and the theory behind speech and phonetics. Our solution generates highly accurate and natural animations of the jaw, lips and tongue through the use of additional phonetic information during the neural network training, and the procedural mapping of its outputs directly to FACS [Prince et al. 2015] based blendshapes, in order to comply to animation industry standards.

KEYWORDS

lipsync, animation, speech animation, FACS, computer vision, neural networks, digital human

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1 INTRODUCTION AND RELATED WORK

The need for realistic facial animations is increasing both in the quality demanded and the number of different sectors looking to employ such a technology. From 3D animation studios to customer service platforms and social media apps, the use of digital humans is becoming more and more popular. The interaction quality is highly dependent on the degree of realism and naturalness the human participant feels throughout the experience, one of the most important factors contributing to this experience being the natural mouth and tongue animations in synchronisation to the speech attributed to the 3D model.

Until a few years ago, the standard methods of animating lips were through facial motion capture, keyframe animation, or early procedural methods. Newer procedural methods ([Charalambous et al. 2019]) struggled to achieve a high level of realism, while other approaches from [Cudeiro et al. 2019], nVidia [Karras et al. 2017] or

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Disney [Taylor et al. 2017] have pipeline constraints which aren't optimal for studios and animators. Jali Research, through [Zhou et al. 2018] and [Edwards et al. 2020] have made significant progress in the area. Our work, inspired by the excellent efforts of our fellow researchers, is looking to improve and raise the standard of the generated animation by focusing on the particular articulatory descriptors corresponding to each phoneme and expressed through subtle differences in the individual FACS units which become building blocks of our animation. Furthermore, our approach generates speech animation by articulating all the visible elements of the speaking apparatus: the jaw, the lips and the tongue.

2 RESULTS

Our work is based on the assumption that each phoneme is better defined in its context by using a combination of articulatory phonetic descriptors, which are tightly correlated with the movements of the jaw, lips and tongue. We selected a number of 6 phonetic descriptors to add as additional features to use throughout the pipeline of LIPSYNC.AI; these are: place of articulation, manner of articulation, is the phoneme voiced, phoneme muscle movement intensity, lexical stress of the phoneme, and phoneme composition.

We developed the LIPSYNC.AI pipeline to take as input an audio file of a speech recording and the transcript as a text file. The first neural network serves as a phoneme classifier and helps create a phonetic timeline which describes what phoneme and for how long it is active in the audio recording. The lip articulation neural network, with a model architecture inspired by [Thangthai et al. 2019], uses this information to predict the less detailed movements of the lips and jaw while correctly simulating the lip coarticulation effect. For each frame it predicts a set of 3 euclidean distances describing the mouth and the jaw positions. The addition of the articulatory phonetic descriptors to the lip articulation model's architecture has improved the final result as seen in Table 1

Table 1: Benchmark results for different LIPSYNC.AI iterations and FaceCap facial motion capture iOS app

	Mandarin Chinese	US English
Human reference	1.0	1.0
LIPSYNC.AI (with art. desc.)	0.81	0.85
LIPSYNC.AI (without art. desc.)	0.75	0.78
FaceCAP ARkit (cleaned animation)	0.83	-

Using the phonetic timeline and the details described through the articulatory phonetic descriptors we add the finer movements to the lips, and create key positions for the tongue corresponding to each phoneme (Figure 2). Our algorithm then interpolates the positions over time, applying specific constraints and corrections to simulate the tongue coarticulation effect and avoid unrealistic rapid movements or mesh collisions.



Figure 2: Tongue rig positions captured during the pronunciation of specific phonemes

At the end of the process, a series of 24 values which will drive blendshapes and rig bones, is output for each frame of the animation.

Currently our solution is available for English, Mandarin Chinese and Arabic MSA languages.

3 LIVE DEMONSTRATION

For SIGGRAPH Asia 2021 on-venue demonstration our technology is presented through a live 1-to-1 conversation between the visiting user and a digital human displayed on the screen.

Using a pipeline which combines other in-house developed technologies, such as Automatic Speech Recognition, alongside a third-party conversational chat bot, we create an interactive experience for our visitors by enabling them to speak with and see real-time feedback from a computer generated human-like entity. The front-end of our pipeline is created in Unreal Engine and uses a MetaHumans 3D avatar with speech animation driven by LIPSYNC.AI. In the case of a purely virtual conference, the on-site setup can be reproduced and executed through a video conference app such as Google Meets or Zoom.

We believe that such an experience offers the visitor a real feeling of an upgraded digital human interaction.

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To Emotech.

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